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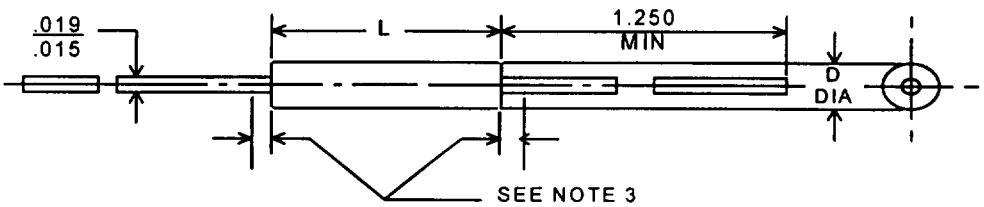
MIL-PRF-39014/21(USAF)
16 August 1976

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE),
ESTABLISHED RELIABILITY,
STYLES CKR31 AND CKR32

This specification sheet is approved for use by the Aeronautical Systems Division, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-PRF-39014.



STYLE	DIMENSIONS			
	L		D	
	INCHES	MM	INCHES	MM
CKR31	.250 Max. .230 Min.	6.35 Max. 5.84 Min.	.100 Max. .080 Min.	2.54 Max. 2.03 Min.
CKR32	.250 Max. .230 Min.	6.35 Max. 5.84 Min.	.140 Max. .120 Min.	3.56 Max. 3.05 Min.

INCHES	MM
.010	.25
.015	.38
.019	.48
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm..
3. Non-conductive material shall not extend beyond .010 (.25 mm) from end of capacitor body.

FSC 5910

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REQUIREMENTS:

Dimensions and configuration: See figure.

Case type: Tubular.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: See table I.

Dissipation factor (DF): See table II.

Insulation resistance (IR):
At 25°C - Not less than 100,000 megohms or 1,000 megohm-microfarads, whichever is less.
At 125°C - Not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.

Insertion loss: Not applicable.

Seal: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-39014. See table II.

Immersion:
DWV - 250 percent of rated voltage.
IR - 10,000 megohms, min or 100 megohm-microfarads, whichever is less.
Capacitance change - See table II.
DF - See table II.

Salt spray (corrosion): Not applicable.

Resistance to soldering heat:
IR - 100,000 megohms, min or 1,000 megohm-microfarads, whichever is less.
Capacitance change - See table II.
DF - See table II.

Life:
Rated conditions, 100 percent of dc rated voltage at 125°C.
0 through 2,000 hours:
IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC - See table II.
DF - See table II.
4,000 through 32,000 hours:
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC - See table II.
DF - See table II.

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Accelerated condition, 200 percent of dc rated voltage at 125°C.
C and 250 hours:
IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC - See table II.
DF - See table II.
1,000, 2,000, and 4,000 hours:
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC - See table II.
DF - See table II.

Part number: MIL-C-39014/21-(dash number from table I).

Marking: In accordance with MIL-C-39014, except that minimum marking as shown in the following example will be permitted. Full marking shall be included on the package.

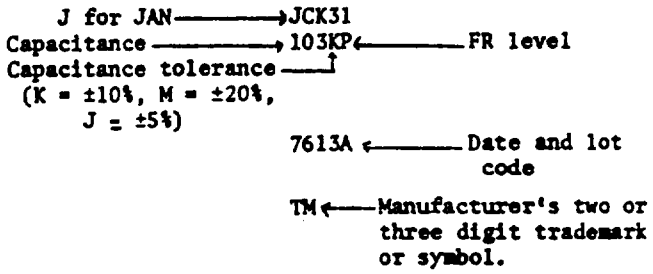


TABLE I. Dash number and electrical characteristics.

Dash number				Capacitance (pF)	Capacitance tolerance symbol	DC rated voltage (volts)
Failure rate level (%/1,000 hours)						
1.0 (M)	0.1 (P)	0.01 (R)	0.001 (S)			
Style CKR31, voltage-temperature limits of +2, -10% and +2, -10%						
0001	0101	Part numbers to be assigned when needed.	Part numbers to be assigned when needed.	270	J	50
0002	0102			270	K	
0003	0103			330	J	
0004	0104			330	K	
0005	0105			390	J	
0006	0106			390	K	
0007	0107			470	J	
0008	0108			470	K	
0009	0109			560	J	
0010	0110			560	K	
0011	0111			680	J	
0012	0112			680	K	
0013	0113			820	J	
0014	0114			820	K	
0015	0115			1000	J	
0016	0116			1000	K	
0017	0117			1200	J	
0018	0118			1200	K	
0019	0119			1500	J	
0020	0120			1500	K	
0021	0121			1800	J	
0022	0122			1800	K	

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TABLE I. Dash number and electrical characteristics.-Continued.

Dash number				Capacitance (pF)	Capacitance tolerance symbol	DC rated voltage (volts)
Failure rate level (%/1,000 hours)						
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR31, voltage-temperature limits of +2, -10% and +2, -10%						
0023	0123	Part numbers to be assigned when needed.	Part numbers to be assigned when needed.	2200	J	50
0024	0124			2200	K	
0025	0125			2700	J	
0026	0126			2700	K	
0027	0127			3300	J	
0028	0128			3300	K	
0029	0129			3900	J	
0030	0130			3900	K	
0031	0131			4700	J	
0032	0132			4700	K	
0033	0133			5600	J	
0034	0134			5600	K	
0035	0135			6800	J	
0036	0136			6800	K	
0037	0137			8200	J	
0038	0138			8200	K	
0039	0139			10000	J	
0040	0140			10000	K	50
Style CKR31, voltage-temperature limits of +2, -15% and +2, -15%						
0041	0141			12,000	K	50
0042	0142			12,000	M	
0043	0143			15,000	K	
0044	0144			15,000	M	
0045	0145			18,000	K	
0046	0146			18,000	M	
0047	0147			20,000	K	
0048	0148			20,000	M	50
Style CKR31, voltage-temperature limits of +20, -45% and +20, -50%						
0049	0149			22,000	K	50
0050	0150			22,000	M	
0051	0151			27,000	K	
0052	0152			27,000	M	
0053	0153			33,000	K	
0054	0154			33,000	M	
0055	0155			39,000	K	
0056	0156			39,000	M	
0057	0157			47,000	K	
0058	0158			47,000	M	
0059	0159			51,000	K	
0060	0160			51,000	M	50

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TABLE I. Dash number and electrical characteristics.-Continued

Dash number Failure rate level (%/1,000 hours)				Capacitance (pF)	Capacitance tolerance symbol	DC rated voltage (volts)
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR32, Voltage-temperature limits of +2, -10% and +2, -10%						
0061	0161			12,000	J	50
0062	0162			12,000	K	
0063	0163			15,000	J	
0064	0164			15,000	K	
0065	0165			18,000	J	
0066	0166			18,000	K	
0067	0167			20,000	J	
0068	0168			20,000	K	50
Style CKR32, Voltage-temperature limits of +2, -15% and +2, -15%						
0069	0169			22,000	K	50
0070	0170			22,000	M	
0071	0171			27,000	K	
0072	0172			27,000	M	
0073	0173			33,000	K	
0074	0174			33,000	M	
0075	0175			39,000	K	
0076	0176			39,000	M	50
Style CKR32, voltage-temperature limits of +20, -45% and +20, -50%						
0077	0177			56,000	K	50
0078	0178			56,000	M	
0079	0179			62,000	K	
0080	0180			62,000	M	
0081	0181			68,000	K	
0082	0182			68,000	M	
0083	0183			75,000	K	
0084	0184			75,000	M	
0085	0185			82,000	K	
0086	0186			82,000	M	
0087	0187			91,000	K	
0088	0188			91,000	M	
0089	0189			100,000	K	
0090	0190			100,000	M	50

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TABLE II.

Capacitance change with reference to 25°C		Dissipation factor (max)	Capacitance change	
Steps A through D of table V Bias = 0 volts	Steps E through G of table V Bias = Rated voltage		After immersion, moisture resistance, and resistance to soldering heat	After life
+2, -10%	+2, -10%	1.0%	±2%	±5%
+2, -15%	+2, -15%	1.5%	±2%	±10%
+20, -45%	+20, -50%	3.0%	±5%	±20%

Custodian:
Air Force - 11

Review activities:
Air Force - 11, 17, 85, 99
DSA - ES

Preparing activity:
Air Force - 11

Agent:
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